

Regression model

A regression model fits a chosen function family to observed data.

Must every observed point lie on the fitted curve?

Equal steps and ratios

At $t = 0, 1, 2, 3$ hours, the respective amounts are 20, 30, 45, 67.5 grams.

Do the data suggest an exponential model?

Exact fit versus regression

At $t = 0, 1, 2, 3$ hours, measured amounts are 20.1, 30.3, 44.7, 68.0 grams.

Why might regression be more appropriate than forcing a curve through two selected points?

An exponential regression

At $t = 0, 1, 2, 3$ hours, $y = 20.1, 30.3, 44.7, 68.0$ grams, respectively.

$$y = ab^t$$

What exponential model does your graphing tool fit to these data?

Interpreting fitted parameters

V is equipment value in dollars and t is elapsed years, $t \geq 0$.

$$V(t) = 950(0.82)^t$$

What initial value and annual percent decrease does the fitted model estimate?

Predicting with a fitted model

Q is mass in grams and t is elapsed hours. The data covered 0 through 3 hours.

$$Q(t) = 19.9(1.505)^t$$

What amount does the model predict at 2.5 hours, to the nearest tenth of a gram?

A residual

The observed mass at 2 hours is 44.7 grams. Define residual as observed minus predicted.

$$Q(t) = 19.9(1.505)^t$$

What is the residual at 2 hours, to the nearest tenth of a gram?

Residual patterns

Residuals are observed minus predicted; the data increase rapidly at first and then level off.

What does a persistent curved residual pattern suggest about a fitted linear model?

Logarithmic regression

The pairs (x, y) are $(1, 2.1)$, $(2, 4.0)$, $(4, 6.3)$, and $(8, 8.2)$.

$$y = a + b \ln x$$

What logarithmic model does your graphing tool fit to these data?

A logarithmic model domain

$$R(x) = 2 + 3\ln x$$

Which input values are allowed?

Prediction from a logarithmic fit

The fitted data used inputs from 1 through 8.

$$R(x) = 2.05 + 2.97 \ln x$$

What does the model predict at $x = 3$, rounded to two decimal places?

Logistic regression

At $t = 0, 1, 2, 3, 4$ weeks, $P = 50, 85, 131, 179, 219$, respectively.

$$P(t) = \frac{L}{1 + Ae^{-kt}}$$

What logistic model does your graphing tool fit to these data?

Interpreting a logistic fit

P is a population count and t is elapsed weeks, $t \geq 0$.

$$P(t) = \frac{300}{1 + 5e^{-0.7t}}$$

What upper limiting population does this fitted model estimate?

Three model families

Each unit increase in input multiplies the positive output by approximately the same factor. Consider exponential, logarithmic, and logistic models.

Which model family best represents these features?

Interpolation and extrapolation

The fitted measurements covered $t = 0$ through 3 hours.

$$Q(t) = 19.9(1.505)^t$$

Which prediction is interpolation: $t = 2.5$ hours or $t = 12$ hours?

A prediction needs context

An exponential model fits measurements collected during the first three hours of an experiment.

What additional evidence would support using the fitted curve far beyond the observed interval?